CHAPTER I

INTRODUCTION

1.1 Overview

To understand a programming language you must practice the programs, this way you can learn the language faster.

Java was a originally developed by **James Gosling** at Sun Microsystems and released in 1995. It is a general-purpose programming language that is class-based and object-oriented. Its motto is "write once, run anywhere", meaning that compiled Java code can run on all platforms that supports Java. There are lots of applications and websites that will not work unless you have Java installed, and more are created every day. Java is fast, secure, and reliable.

Java is freely downloadable. The Java Runtime Environment **(JRE)** is what you get when you download Java software. The JRE consists of the Java Virtual Machine**(JVM)**, Java platform core classes, and supporting Java platform libraries. The JRE is the runtime portion of Java software, which is all you need to run it in your Web browser.

Java platform is a collection of programs that help to develop and run programs written in the Java programming language. Java platform includes an execution engine, a compiler, and a set of libraries. JAVA is **platform-independent language**. It is not specific to any processor or operating system.

Java Development Kit (JDK) consists of compiler which takes the Java program as input and generates the Java byte code as output.

The **latest version of Java** is **Java** SE 13.0.1 released on September **2019**.

1.2 Java Getting Started

We need to install Java SE Development Kit(JDK and SE means Standard Edition)and it is freely downloadable from oracle.com.

To install Java on Windows:

1. Go to "System Properties" (Can be found on Control Panel > System and Security > System > Advanced System Settings)

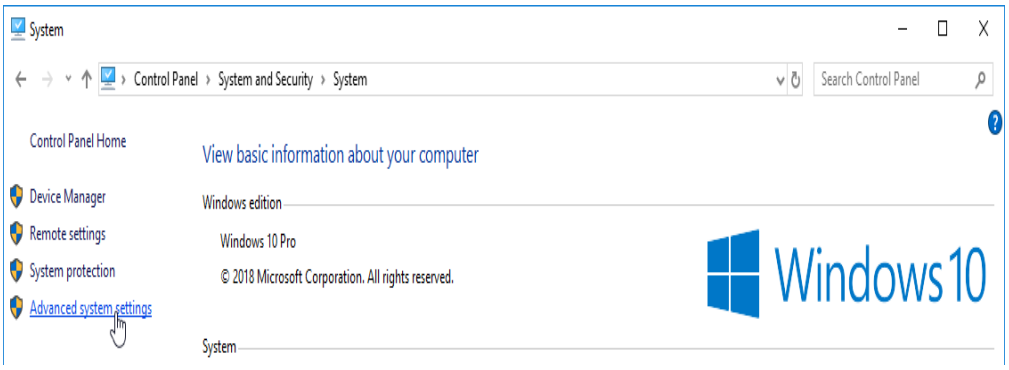
2. Click on the "Environment variables" button under the "Advanced" tab

3. Then, select the "Path" variable in System variables and click on the "Edit" button

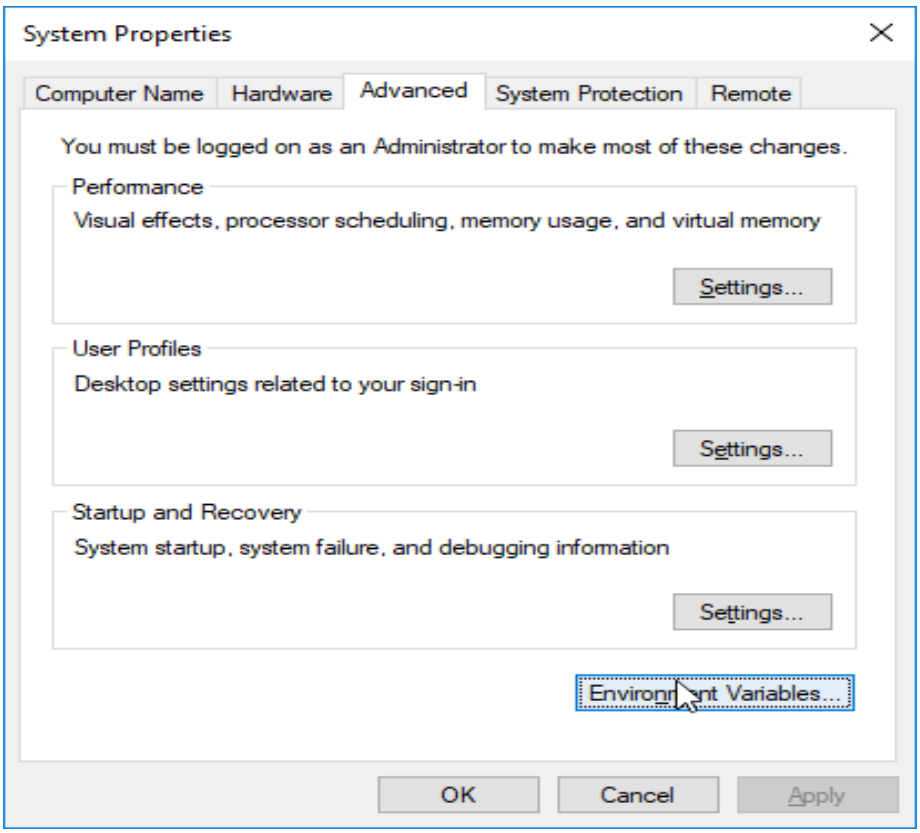
4. Click on the "New" button and add the path where Java is installed, followed by **\bin**. By default, Java is installed in C:\Program Files\Java\jdk-11.0.1 (If nothing else was specified when you installed it). In that case, You will have to add a new path with: **C:\Program Files\Java\jdk-11.0.1\bin.** Then, click "OK", and save the settings

5. At last, open Command Prompt (cmd.exe) and type **java -version** to see if Java is running on your machine

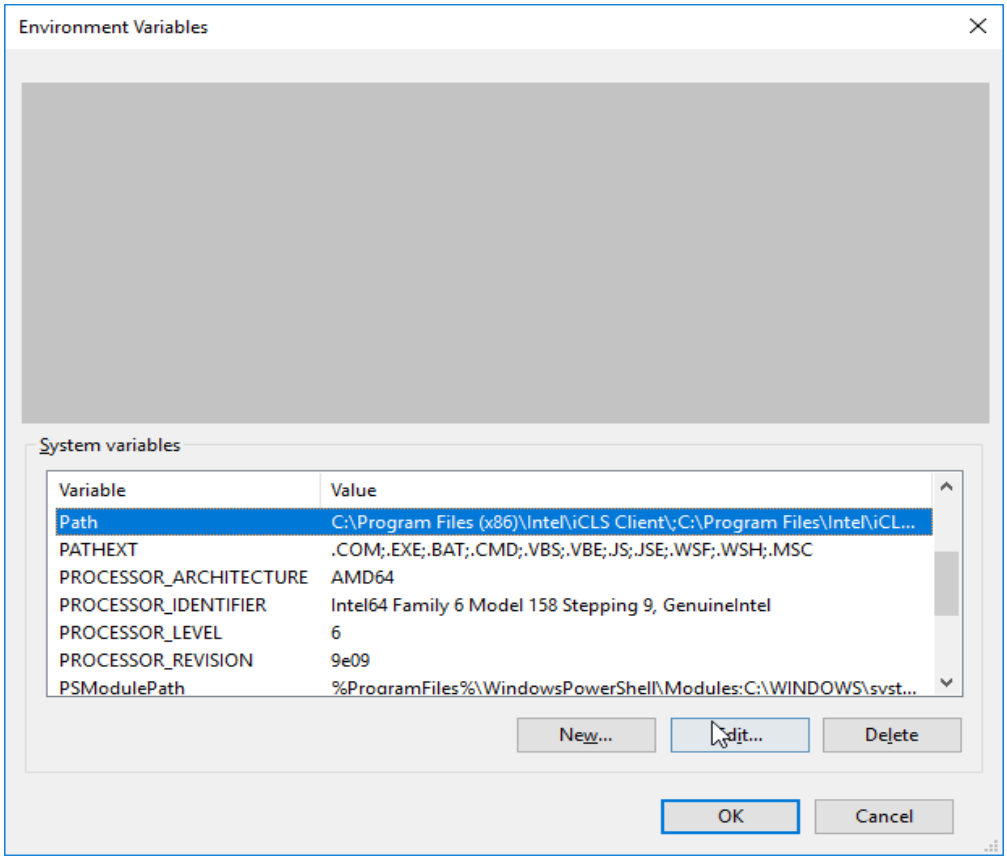
Step: 1



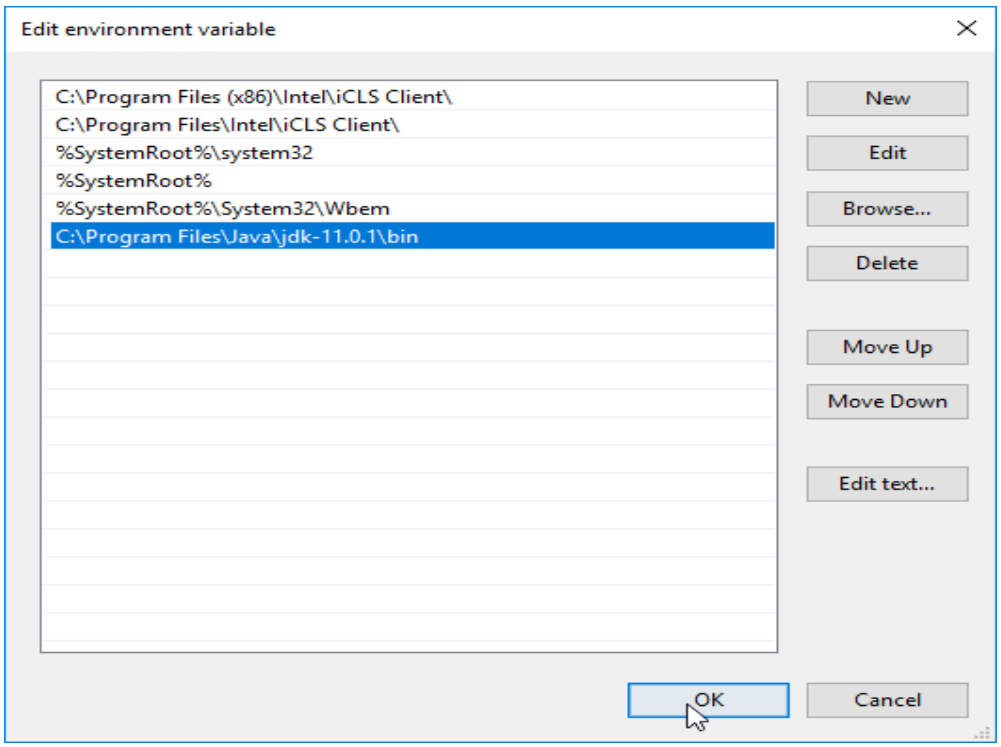
Step: 2



Step: 3


Environment Variables
System variables
Variable
Value
Path
C:\Program Files (x86)\Intel\iCLS Client\;C:\Program Files\Intel\iCL...
PATHEXT
.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC
PROCESSOR_ARCHITECTURE
AMD64
PROCESSOR_IDENTIFIER
Intel64 Family 6 Model 158 Stepping 9, GenuineIntel
PROCESSOR_LEVEL
6
PROCESSOR_REVISION
9e09
PSModulePath
%ProgramFiles%\WindowsPowerShell\Modules;C:\WINDOWS\syst...
New...
Edit...
Delete
OK
Cancel

Step: 4


Edit environment variable
C:\Program Files (x86)\Intel\iCLS Client\
C:\Program Files\Intel\iCLS Client\
%SystemRoot%\system32
%SystemRoot%
%SystemRoot%\System32\Wbem
C:\Program Files\Java\jdk-11.0.1\bin
New
Edit
Browse...
Delete
Move Up
Move Down
Edit text...
OK
Cancel

Step: 5

Write the following in the command line (cmd.exe):

C:\Users*Your Name*>java -version

If Java was successfully installed, you will see something like this (depending on version):

java version "11.0.1" 2018-10-16 LTS
Java(TM) SE Runtime Environment 18.9 (build 11.0.1+13-LTS)
Java HotSpot(TM) 64-Bit Server VM 18.9 (build 11.0.1+13-LTS, mixed mode)

After the installation of the Java software, write the Java code in a notepad. The extension to save the Java program is ***filename.java.*** The file name of the Java program should be same as the class name and Java is a case-sensitive language. Hence **sample.java** and **Sample.java** are two different files.

Java developer should be familiar with:

- javac.exe: is Java compiler that translates programs written in Java code into bytecode form.
- java.exe: is the Java Virtual Machine launcher that executes bytecode.

1.3 Simple Java Problems

Step 1: Open a simple text editor program such as Notepad and type the following content:

```
import java.io.*;
public class WelcomeJava
{
public static void main(String[] args)
{
System.out.println("Welcome to Java");
}
}
```

Step 2: Save the file as WelcomeJava.java (note that the extension is .java) under a directory, let's say, *C:\jdk1.3\bin*

Step 3: Go to Command prompt. Type the following command to change the current directory to the one where the source file is stored:

cd C:\jdk1.3\bin

And type the following command to compile:

javac WelcomeJava.java

Step 4: Type the following command to run:

java WelcomeJava

Output: Welcome to Java

CHAPTER II

BLOCK I: JAVA FUNDAMENTAL PROBLEMS

2.1 SIMPLE JAVA PROBLEMS:

1. Program to Add Two Integers

```
import java.io.*;
public class AddTwoIntegers
{
  public static void main(String[] args)
{
        int first = 10;
        int second = 20;
        int sum = first + second;
        System.out.println("The sum is: " + sum);
    }
}
```

Output:

```
C:\jdk1.3\bin>javac AddTwoIntegers.java
C:\jdk1.3\bin>java AddTwoIntegers
The sum is: 30
```

2. Program to Find the Greatest of three Numbers

```
import java.io.*;
import java.io.*;
class Greatest
{
public static void main(String args[])
{
int a,b,c;
try
{
DataInputStream d= new DataInputStream(System.in);
```

```
System.out.println("enter the value of a:");
a=Integer.parseInt(d.readLine());
System.out.println("enter the value of b:");
b=Integer.parseInt(d.readLine());
System.out.println("enter the value of c:");
c=Integer.parseInt(d.readLine());
if(a>b)
{
System.out.println("greatest no is a="+a);
}
else if(c>b)
{
System.out.println("greatest no is c="+c);
}
else
{
System.out.println("greatest no is b="+b);
}
}
catch (Exception e)
{
}
}
}
```

Output:

```
C:\jdk1.3\bin>javac Greatest.java
C:\jdk1.3\bin>java Greatest
enter value of a:
12
enter value of b:
25
enter value of c:
10
greatest no is b=25
```

3. Program to Find the Factorial of a number using for loop

```
import java.io.*;
public class Factorial
{
   public static void main(String[] args)
{
       int num = 5;
       long factorial = 1;
       for(int i = 1; i <= num; ++i)
       {
          factorial *= i;
       }
       System.out.printf("Factorial of %d = %d", num, factorial);
   }
}
```

Output:

Factorial of 5 = 120

4. Program to swap two numbers

```
import java.io.*;
public class SwapNumbers
{
public static void main(String[]args)
{
float a=3.4f, b=7.2f;
System.out.println("—Beforeswap--");
System.out.println("Firstnumber="+a);
System.out.println("Secondnumber="+b);
float t=a; a= b;
b=t;
System.out.println("--Afterswap--");
System.out.println("Firstnumber="+a);
System.out.println("Secondnumber="+b);
}
}
```

Output:

C:\jdk1.3\bin>javac SwapNumbers.java

C:\jdk1.3\bin>java SwapNumbers

--Before swap—

First number =3.4

Second number = 7.2

--After swap—

First number = 7.2

Second number = 3.4

2.2 Class and Objects:

5. Write a Java Program to define a Class and its Methods

```
/Java Program to demonstrate the working of a banking-system
//where we deposit and withdraw amount from our account.
//Creating an Account class which has deposit() and withdraw() methods
class Account{
int acc_no;
String name;
float amount;
//Method to initialize object
void insert(int a,String n,float amt)
{
acc_no=a;
name=n;
amount=amt;
}
//deposit method
void deposit(float amt)
{
amount=amount+amt;
System.out.println(amt+" deposited");
}
```

```
//withdraw method
void withdraw(float amt)
{
if(amount<amt)
{
System.out.println("Insufficient Balance");
}
else
{
amount=amount-amt;
System.out.println(amt+" withdrawn");
}
}
//method to check the balance of the account
void checkBalance(){System.out.println("Balance is: "+amount);
}
//method to display the values of an object
void display(){System.out.println(acc_no+" "+name+" "+amount);
}
}
//Creating a test class to deposit and withdraw amount
class TestAccount
{
public static void main(String[] args)
{
Account a1=new Account();
a1.insert(832345,"Ankit",1000);
a1.display();
a1.checkBalance();
a1.deposit(40000);
a1.checkBalance();
a1.withdraw(15000);
a1.checkBalance();
```

```
 }
 }
```

Output:

```
C:\jdk1.3\bin>javac TestAccount.java
C:\jdk1.3\bin>java TestAccount
832345   Ankit      1000.0
Balance is: 1000.0
40000.0  deposited
Balance is: 41000.0
15000.0   withdrawn
Balance is: 26000.0
```

6. Write a Java Program for Constructor Overloading

```
import java.io.*;
class Student
{
   int id;
   String name;
   int age;

   //creating two arg constructor
   Student(int i, String n)
  {
   id = i;
   name = n;
   }
   //creating three arg constructor
   Student(int i, String n, int a)
  {
   id = i;
   name = n;
   age=a;
   }
   void display()
  {
  System.out.println(id+" "+name+" "+age);
  }

   public static void main(String args[])
```

```
{
  Student s1 = new Student(101,"Ashwath");
  Student s2 = new Student(102,"Yazhini",25);
  s1.display();
  s2.display();
}
}
```

Output:

```
C:\jdk1.3\bin>javac Student.java
C:\jdk1.3\bin>java Student
101   Ashwath  0
102   Yazhini   25
```

7. Program using String Class and its Methods

```
import java.lang.String;
class stringdemo
{
public static void main(String args[])
{
String s1=new String("distance education");
String s2=new String ("DISTANCE EDUCATION");
System.out.println(" The string s1 is : " +s1);
System.out.println(" The string s2 is : " +s2);
System.out.println(" Length of the string s1 is : " +s1.length());
System.out.println(" The first occurrence of u is at the position : "
+s1.indexOf('t'));
System.out.println(" The String in Upper Case : " +s1.toUpperCase());
System.out.println(" The String in Lower Case : " +s2.toLowerCase());
System.out.println(" s1 equals to s2 : " +s1.equals(s2));
System.out.println(" s1 equals ignore case to s2 : " +s1.equalsIgnoreCase(s2));
System.out.println(" Character at an index of 6 is :" +s1.charAt(6));
String s3=s1.substring(10,12);
System.out.println(" Extracted substring is :"+s3);
System.out.println(" After Replacing a with g in s1 : " + s1.replace('i','o'));
```

```
String s4="    This is a book    ";
System.out.println(" The string s4 is :"+s4);
System.out.println(" After trim() :"+s4.trim());
}
}
```

Output:

```
c:\jdk1.3\bin>javac stringdemo.java
c:\jdk1.3\bin>java stringdemo

The strings1 is :distance education
The strings2 is :DISTANCE EDUCATION
Length of the strings1 is : 18
The first occurrence of u is at the position: 3
The String in UpperCase: DISTANCE EDUCATION
The String in LowerCase: distance education
s1equals to s2: false
s1equals ignore case to s2: true
Character at an index of 6 is : c
Extracted substring is : edu
After Replacing i with o in s1: dostance educatoon
The strings4 is:   This is a book
After trim():This is a book
```

8. Program using StringBuffer Class and its Methods

```
import java.lang.String;
class stringbufferdemo
{
public static void main(String args[])
{
StringBuffer sb=new StringBuffer("Alagappa University");
System.out.println("This string in sb is : " +sb);
System.out.println("The length of the string sb is : " +sb.length());
```

```
System.out.println("The capacity of the string sb is : " +sb.capacity());
System.out.println("The character at an index of  is : " +sb.charAt(7));
sb.setCharAt(3,'x');
System.out.println("After setting char x at position 3 : " +sb);
System.out.println("After appending : " +sb.append(" is in karaikudi "));
System.out.println("After inserting : " +sb.insert(20,"is "));
System.out.println("After deleting : " +sb.delete(20,22));
}
}
```

Output:

```
c:\jdk1.3\bin>javac stringbufferdemo.java
c:\jdk1.3\bin>java stringbufferdemo
This string in sb is : Alagappa University
The length of the string sb is : 19
The capacity of the string sb is : 35
The character at an index of 7 is : a
After setting char x at position 3 : Alaxappa University
After appending : Alagappa University in karaikudi
After inserting : Alagappa University is in karaikudi
After deleting : Alagappa University in karaikudi
```

9. Program using Wrapper Class

Autoboxing

```
//Java program to convert primitive into objects
//Autoboxing example of int to Integer
public class WrapperExample1
{
public static void main(String args[])
{
//Converting int into Integer
int a=20;
Integer i=Integer.valueOf(a);//converting int into Integer explicitly
```

```
Integer j=a;//autoboxing, compiler will write Integer.valueOf(a) internally Syst
em.out.println(a+" "+i+" "+j);
}
}
```

Output:

20 20 20

Unboxing

```
//Java program to convert object into primitives
//Unboxing example of Integer to int
public class WrapperExample2
{
public static void main(String args[])
{
//Converting Integer to int
Integer a=new Integer(3);
int i=a.intValue();      //converting Integer to int explicitly
int j=a;          //unboxing, now compiler will write a.intValue() internally
System.out.println(a+" "+i+" "+j);
}
}
```

Output:

3 3 3

2.3 Conditional Control in Java

10. Program to implement If Statement

```
//Java Program to demonstrate the use of Nested If Statement.
public class NestedIf
{
public static void main(String[] args)
{
   //Creating two variables for age and weight
   int age=25;
```

```
   int weight=48;
   //applying condition on age and weight
   if(age>=18)
   {
      if(weight>50)
         {
         System.out.println("You are eligible to donate blood");
         }
         else
         {
           System.out.println("You are not eligible to donate blood");
         }
   }
         else
         {
           System.out.println("Age must be greater than 18");
         }
}  }
```

Output:

C:\jdk1.3\bin>javac NestedIf.java

C:\jdk1.3\bin>java NestedIf

You are not eligible to donate blood

11. Program to implement Switch Statement

```
import java.util.Scanner;
class Calculator {
   public static void main(String[] args) {
   char operator;
            Double number1, number2, result;
            Scanner scanner = new Scanner(System.in);
            System.out.print("Enter operator (either +, -, * or /): ");
```

```
        operator = scanner.next().charAt(0);
        System.out.print("Enter number1 and number2 respectively: ");
        number1 = scanner.nextDouble();
        number2 = scanner.nextDouble();
        switch (operator)
        {
         case '+':
        result = number1 + number2;
        System.out.print(number1 + "+" + number2 + " = " + result);
        break;

         case '-':
        result = number1 - number2;
        System.out.print(number1 + "-" + number2 + " = " + result);
        break;

        case '*':
        result = number1 * number2;
        System.out.print(number1 + "*" + number2 + " = " + result);
        break;

        case '/':
        result = number1 / number2;
        System.out.print(number1 + "/" + number2 + " = " + result);
        break;

        default:
        System.out.println("Invalid operator!");
        break;
        }
    }
}
```

Output:

```
C:\jdk1.3\bin>javac   Calculator.java
C:\jdk1.3\bin>java Calculator
Enter operator (either +, -, * or /):  *
Enter number1 and number2 respectively: 1.4    -5.0
1.4*-5.0 = -7.0
```

2.4 Looping in Java

12. Matrix addition using for loop

```
import java.util.Scanner;

class AddTwoMatrix
{
  public static void main(String args[])
  {
    int m, n, c, d;
    Scanner in = new Scanner(System.in);

    System.out.println("Enter the number of rows and columns of matrix");
    m = in.nextInt();
    n = in.nextInt();

    int first[][] = new int[m][n];
    int second[][] = new int[m][n];
    int sum[][] = new int[m][n];

    System.out.println("Enter the elements of first matrix");

    for (c = 0; c < m; c++)
      for (d = 0; d < n; d++)
        first[c][d] = in.nextInt();

    System.out.println("Enter the elements of second matrix");

    for (c = 0 ; c < m; c++)
      for (d = 0 ; d < n; d++)
        second[c][d] = in.nextInt();

    for (c = 0; c < m; c++)
      for (d = 0; d < n; d++)
        sum[c][d] = first[c][d] + second[c][d];  //replace '+' with '-' to subtract
matrices
    System.out.println("Sum of the matrices:");
    for (c = 0; c < m; c++)
    {
      for (d = 0; d < n; d++)
      System.out.print(sum[c][d] + "\t");
```

```
      System.out.println();
    }
  }
}
```

Output:

```
E:\Java>javac AddTwoMatrix.java

E:\Java>java AddTwoMatrix
Enter the number of rows and columns of matrix
2 2
Enter the elements of first matrix
1 2
3 4
Enter the elements of second matrix
4 3
2 1
Sum of entered matrices:-
5       5
5       5

E:\Java>
```

13. Program to check Palindrome Number using while loop

```
import java.io.*;
class PalindromeExample
{
 public static void main(String args[])
{
  int r,sum=0,temp;
  int n=454;     //It is the number variable to be checked for palindrome
   temp=n;
  while(n>0)
  {
   r=n%10;       //getting remainder
   sum=(sum*10)+r;
   n=n/10;
  }
  if(temp==sum)
   System.out.println("Palindrome number ");
  else
   System.out.println("not palindrome");
}}
```

Output:

Palindrome number

CHAPTER III

BLOCK II: OOP CONCEPTS

3.1 METHOD OVERLOADING

Overloading allows different methods to have the same name, but different signatures where the signature can differ by the number of input parameters or type of input parameters or both. Overloading is related to compile-time (or static) polymorphism.

14. Program to find the area of different shapes using Method Overloading

```
import java.io.*;
class OverloadDemo
{
   void area(float x)
   {
   System.out.println(The area of the square is "+Math.pow(x, 2)+" sq units");
   }
   void area(float x, float y)
   {
    System.out.println("The area of the rectangle is "+x*y+" sq units");
   }
   void area(double x)
   {
    double z = 3.14 * x * x;
    System.out.println("The area of the circle is "+z+" sq units");
   }
}
class Overload
{
    public static void main(String args[])
   {
      OverloadDemo ob = new OverloadDemo();
      ob.area(5);
      ob.area(11,12);
      ob.area(2.5);
       }
}
```

Output:

```
C:\jdk1.3\bin>javac OverloadDemo.java
C:\jdk1.3\bin>java OverloadDemo

The area of the square is 25.0 sq units
The area of the rectangle is 132.0 sq units
The area of the circle is 19.625 sq units
```

15. Method Overloading by changing the data type of parameters

```
import java.io.*;
class MethodOverloading {

   // this method accepts int
   private static void display(int a){
      System.out.println("Got Integer data.");
   }

   // this method  accepts String object
   private static void display(String a){
      System.out.println("Got String object.");
   }

   public static void main(String[] args) {
      display(1);
      display("Hello");
   }
}
```

Output:

```
C:\jdk1.3\bin>javac MethodOverloading.java
C:\jdk1.3\bin>java MethodOverloading

Got Integer data.
Got String object.
```

Note: Java does not support operator overloading

3.2 INHERITANCE AND PACKAGES

Inheritance in Java is a mechanism in which one object acquires all the properties and behaviors of a parent object. In object oriented programming, inheritance is used to promote the code re-usability.

16. Program to demonstrate Single Inheritance

```
import java.io.*;
class Calculation{
int z;
public void addition(int x, int y)
{
z=x+y;
System.out.println("The sum of the given numbers:"+z);
}
public void Substraction(int x,int y)
{
z=x-y;
System.out.println("The difference between the given numbers:"+z);
}
}
public class My_Calculation extends Calculation
{
public void multiplication(int x, int y)
{
z=x*y;
System.out.println("The product of the given numbers:"+z);
}
public static void main(String args[])
{
 int a=20, b=10;
 My_Calculation demo = new My_Calculation();
 demo.addition(a, b);
 demo.Substraction(a, b);
 demo.multiplication(a, b);
}
}
```

Output:

```
C:\jdk1.3\bin>javac My_Calculation.java
C:\jdk1.3\bin>java My_Calculation
The sum of the given numbers:30
The difference between the given numbers:10
The product of the given numbers:200
```

18. Program to calculate employee salary using Hierarchical Inheritance with 'super' keyword

```
import java.util.Scanner;
class Employee {
int Emp_id;
String Emp_name;
String Address;
String Mail_Id;
String Mobile_no;
Employee(){
}
Employee(int id,String name,String addr,String mail,String mob)
{
 this.Emp_id=id;
 this.Emp_name=name;
 this.Address=addr;
 this.Mail_Id=mail;
 this.Mobile_no=mob;
}
}
class Programmer extends Employee
{
double BP, Gross_salary, Net_salary;
public Programmer(int id,String name,String addr,String mail,String mob)
{
super(id,name,addr,mail,mob);
}
void computePay( )
{
System.out.print("Enter Basic Pay:");
Scanner input=new Scanner(System.in);
BP=input.nextDouble( );
double DA,HRA,PF,Fund;
```

```
DA=(BP*97/100);
HRA=(BP*10/100);
PF=(BP*12/100);
Fund=(BP*0.1/100);
Gross_salary=BP+DA+HRA;
Net_salary=BP+DA+HRA-(PF+Fund);
System.out.println("Emp_ID: "+Emp_id);
System.out.println("Emp_Name: "+Emp_name);
System.out.println("Address: "+Address);
System.out.println("Mail_ID: "+Mail_Id);
System.out.println("Mobile Number: "+Mobile_no);
System.out.println("Gross Pay: "+Gross_salary);
System.out.println("Net Pay: "+Net_salary);
}
}
class Professor extends Employee
{
double BP,Gross_salary,Net_salary;
public Professor(int id,String name,String addr,String mail,String mob)
{
super(id,name,addr,mail,mob);
}
 void computePay( )
{
System.out.print("Enter Basic Pay:");
Scanner input=new Scanner(System.in);
BP=input.nextDouble();
Gross_salary=BP;
double DA,HRA,PF,Fund;
DA=(BP*97/100);
HRA=(BP*10/100);
PF=(BP*12/100);
Fund=(BP*0.1/100);
```

```
Net_salary=BP+DA+HRA-(PF+Fund);
System.out.println("Emp_ID: "+Emp_id);
System.out.println("Emp_Name: "+Emp_name);
System.out.println("Address: "+Address);
System.out.println("Mail_ID: "+Mail_Id);
System.out.println("Mobile Number: "+Mobile_no);
System.out.println("Gross Pay: "+Gross_salary);
System.out.println("Net Pay: "+Net_salary);
}
}
public class PaySlip
{
public static void main(String[] args)
{
Programmer p=new
        Programmer(10,"AAA","xxx","aaa_xxx@gmail.com","1111111111");
System.out.println("------- Programmer ----------");
p.computePay( );
Professor Ap=new
        Professor(20,"BBB","yyy","bbb_yyy@gmail.com","2222222222");
System.out.println("------- Assistant Professor ----------");
Ap.computePay( );
}
}
```

Output:

```
C:\jdk1.3\bin>javac PaySlip.java
C:\jdk1.3\bin>java PaySlip
------- Programmer ----------
Enter Basic Pay:5000
Emp_ID: 10
Emp_Name: AAA
Address: xxx
Mail_ID: aaa_xxx@gmail.com
Mobile Number: 1111111111
```

```
Gross Pay: 10350.0
Net Pay: 9745.0
------- Assistant Professor ----------
 Enter Basic Pay:10000
Emp_ID: 20
Emp_Name: BBB
Address: yyy
Mail_ID: bbb_yyy@gmail.com
Mobile Number: 2222222222
Gross Pay: 10000.0
Net Pay: 19490.0
```

19. Write a program to demonstrate the use of Interface

In Java, multiple inheritance is achieved through Interface.

```
import java.io.*;
interface Drawable
{
  int RED = 1;
  int GREEN = 2;
  int BLUE = 3;
  int BLACK = 4;
  int WHITE = 5;
  void draw(int color);
}
class Circle implements Drawable
{
  private double x, y, radius;
  Circle(double x, double y, double radius)
  {
    this.x = x;
    this.y = y;
    this.radius = radius;
  }
  @Override
  public void draw(int color)
  {
    System.out.println("Circle drawn at (" + x + ", " + y + "), with radius " +
radius + ", and color " + color);
  }
```

```
  double getRadius()
  {
    return radius;
  }
  double getX()
  {
    return x;
  }
  double getY()
  {
    return y;
  }
}
Class Draw
{
public static void main(String[] args)
{
  Circle c = new Circle(10, 20, 15);
  c.draw(Drawable.RED);
}
}
```

Output:

C:\jdk1.3\bin>javac Draw.java

C:\jdk1.3\bin>java Draw

Circle drawn at (30.0, 20.0), with radius 10.0, and color 1

20. Program to demonstrate Multiple Inheritance

```
import java.io.*;
interface MotorBike
{
int speed=50;
public void totalDistance();
}
interface Cycle
{
int distance=150;
public void speed();
}
```

```
public class TwoWheeler implements MotorBike, Cycle
{
int totalDistance;
int avgSpeed;
public void totalDistance()
{
totalDistance=speed*distance;
System.out.println("Total Distance Travelled : "+totalDistance);
}
public void speed()
{
int avgSpeed=totalDistance/speed;
System.out.println("Average Speed maintained : "+avgSpeed);
}
public static void main(String args[])
{
TwoWheeler t1=new TwoWheeler();
t1.totalDistance();
t1.speed();
}
}
```

Output:

```
C:\jdk1.3\bin>javac TwoWheeler.java
C:\jdk1.3\bin>java TwoWheeler
Total Distance Travelled : 7500
Average Speed maintained : 150
```

21. Program using Packages

```
import java.io.*;
package MyPackage;
public class Compare {
  int num1, num2;
  Compare(int n, int m) {
     num1 = n;
     num2 = m;
}
public void getmax()
{
    if ( num1 > num2 )
{
        System.out.println("Maximum value of two numbers is " + num1);
```

```
  }
   else
    {
      System.out.println("Maximum value of two numbers is " + num2);
    }
  }
  public static void main(String args[])
   {
      Compare current[] = new Compare[3];
      current[1] = new Compare(5, 10);
      current[2] = new Compare(123, 120);
          for(int i=1; i < 3 ; i++)
           {
           current[i].getmax();
           }
   }
}
```

Output:

```
C:\jdk1.3\bin>MyPackage> javac Compare.java
C:\jdk1.3\bin>MyPackage> java Compare

Maximum value of two numbers is 10
Maximum value of two numbers is 123
```

```
// importing MyPackage

package Alagappa;
import MyPackage.Compare;
public class Demo{
   public static void main(String args[]) {
      int n=10, m=10;
      Compare current = new Compare(n, m);
      if(n != m) {
          current.getmax();
      }
      else {
         System.out.println("Both the values are same");
      }
} }
```

Output:

C:\jdk1.3\bin>Alagappa>javac Demo.java

C:\jdk1.3\bin>Alagappa>java Demo

Both the values are same

22. Program to implement various access controls to its data members and methods.

```
import java.io.DataInputStream;
class Student
{
Private int rollno;
private String name;
DataInputStream dis=new DataInputStream(System.in);
public void getrollno()
{
try
{
System.out.println("Enter rollno");
rollno=Integer.parseInt(dis.readLine());
System.out.println("Enter name ");
name=dis.readLine();
}
catch(Exception e){ } }
voidputrollno()
{
System.out.println("Roll No ="+rollno);
System.out.println("Name ="+name);
}
}
class Marks extends Student {
protectedint m1,m2,m3;
voidgetmarks()
{
```

```
try
{
System.out.println("Enter marks :");
m1=Integer.parseInt(dis.readLine());
m2=Integer.parseInt(dis.readLine());
m3=Integer.parseInt(dis.readLine());
}
catch(Exception e) { } }
voidputmarks() {
System.out.println("m1="+m1);
System.out.println("m2="+m2);
System.out.println("m3="+m3);
} }
class Result extends Marks
{
private float total;
voidcompute_display()
{
total=m1+m2+m3;
System.out.println("Total marks :" +total);
} }
classMultilevelDemo
{
public static void main(String args[])
{
Result r=new Result();
r.getrollno();
r.getmarks();
r.putrollno();
r.putmarks(); r.compute_display();
}
}
```

Output:

```
C:\jdk1.3\bin>javacMultilevelDemo.java
C:\jdk1.3\bin>javaMultilevelDemo
Enter rollno12345
EnternameAvinash
Entermarks :
54
78
46
RollNo=12345
Name=Avinash
m1=54
m2=78
m3=46
Totalmarks: 178.0
```

3.3 Polymorphism and Message Passing

Polymorphism in Java is a concept by which we can perform a *single action in different ways*. Polymorphism is derived from 2 Greek words: poly and morphs. The word "poly" means many and "morphs" means forms. So polymorphism means many forms.

There are two types of polymorphism in Java: compile-time polymorphism and runtime polymorphism. We can perform polymorphism in java by method overloading and method overriding.

We have already discussed about method overloading a static method in 3.1(Program 14), it is the example of compile time polymorphism. Here, we will focus on runtime polymorphism in java.

23. Program to demonstrate Method Overriding

```
import java.io.*;
class ParentClass
{
  //Parent class constructor
  ParentClass()
  {
  System.out.println("Constructor of Parent");
```

```
   }
   void disp()
  {
   System.out.println("Parent Method");
   }
}
class JavaExample extends ParentClass
{
   JavaExample()
  {
   System.out.println("Constructor of Child");
   }
   void disp( )
  {
   System.out.println("Child Method");
   //Calling the disp() method of parent class
   super.disp();
   }
   public static void main(String args[])
  {
            //Creating the object of child class
   JavaExample obj = new JavaExample();
   obj.disp();
   }
}
```

Output:

```
C:\jdk1.3\bin>javac JavaExample.java
C:\jdk1.3\bin>java JavaExample
Constructor of Parent
Constructor of Child
Child Method
Parent Method
```

24. Message Passing through Methods

```
import java.io.*;
public class MsgPass_Methods
{
void displayInt(int x, int y)
{
int z = x + y;
System.out.println("Int Value is : "+z);
```

```
}
void displayFloat(float x, float y)
{
float z = x * y;
System.out.println("Float Value is : "+z);
}
}
Class MsgPassExample
{
public static void main (String args[])
{
MsgPass_Methodsobj = new MsgPass_Methods();
obj.displayInt(15,40);
obj.displayFloat((float)4.37, (float)2.59);
}
}
```

Output:

```
C:\jdk1.3\bin>javac MsgPassExample.java
C:\jdk1.3\bin>java MsgPassExample

Int Value is : 55
Float Value is : 11.3183
```

CHAPTER IV

BLOCK III: THREAD & VIRTUAL FUNCTION

4.1 Threads

25. Program for thread creation through extending Thread class

```
import java.lang.Thread;
public class MyThread extends Thread
{
    String message;
    MyThread(String msg)
    {
        message=msg;
    }
    public void run()
    {
        for(int count=0;count<=5;count++)
        {
            System.out.println("Run method: " + message);
        }
    }
}
public class MainThread
{
    public static void main(String[] args)
    {
        MyThread t1 = new MyThread("Run");
        MyThread t2 = new MyThread("Thread");
        t1.start();
        t2.start();
    }
}
```

Output:

```
C:\jdk1.3\bin>javac MainThread.java
C:\jdk1.3\bin>java MainThread
Run method: Run
Run method: Run
Run method: Run
Run method: Run
Run method: Run
```

```
Run method: Run
Run method: Thread
Run method: Thread
Run method: Thread
Run method: Thread
Run method: Thread
Run method: Thread
```

26. Program for thread creation through Runnable Interface

```
import java.lang.Runnable;
class MyThread implements Runnable
{
    String message;
    MyThread(String msg)
    {
        message = msg;
    }
    public void run()
    {
        for(int count=0;count<=5;count++)
        {
            System.out.println("Run method: " + message);
        }
    }
}
public class MainMyThread
{
    public static void main(String[] args)
    {
        MyThread dt1=new MyThread("Run");
        MyThread dt2=new MyThread("Thread");
        Thread t1 = new Thread(dt1);
        Thread t2 = new Thread(dt2);
        t1.start();
        t2.start();
    }
}
```

Output:

```
C:\jdk1.3\bin>javac MainMyThread.java
C:\jdk1.3\bin>java MainMyThread
Run method: Thread
Run method: Run
Run method: Thread
```

```
Run method: Run
Run method: Thread
Run method: Run
Run method: Thread
Run method: Thread
Run method: Thread
Run method: Run
Run method: Run
Run method: Run
```

27. Program for Thread Priority

```
import java.io.*;
class Thread1 extends Thread
{
public void run()
{
for(int m=1;m<=5;m++)
System.out.println("\n\tMINIMUM PRIORITY THREAD RUNNING");
System.out.println("\n\tMINIMUM PRIORITY THREAD QUITING");
}
}
class Thread2 extends Thread
{
public void run()
{
for(int n=1;n<=5;n++)
System.out.println("\n\tMAXIMUM PRIORITY THREAD RUNNING");
System.out.println("\n\tMAXIMUM PRIORITY THREAD QUITING");
}
}
class demo
{
public static void main(String args[])
{
Thread1 t1=new Thread1();
t1.setPriority(Thread.MIN_PRIORITY+1);
t1.start();
Thread2 t2=new Thread2();
t2.setPriority(Thread.MAX_PRIORITY-1);
t2.start();
}
}
```

Output:

```
MINIMUM PRIORITY THREAD RUNNING
MINIMUM PRIORITY THREAD RUNNING
MINIMUM PRIORITY THREAD RUNNING
MINIMUM PRIORITY THREAD RUNNING
MINIMUM PRIORITY THREAD RUNNING
MINIMUM PRIORITY THREAD RUNNING
MAXIMUM PRIORITY THREAD RUNNING
MAXIMUM PRIORITY THREAD RUNNING
MAXIMUM PRIORITY THREAD RUNNING
MAXIMUM PRIORITY THREAD RUNNING
MAXIMUM PRIORITY THREAD RUNNING
MAXIMUM PRIORITY THREAD RUNNING
```

Note: Java does not support virtual function

CHAPTER V

BLOCK IV: IO & EXCEPTION HANDLING PROGRAMS

5.1 Exception handling programs

28. Program to implement the concept of Predefined Exception Handling

```
import java.io.*;
class ExceptionPre
{
public static void main(String[] args)
{
    try
   {
      int divideByZero = 5 / 0;
    }
catch (ArithmeticException e)
{
   System.out.println("ArithmeticException => " + e.getMessage());
 }
finally
    {
      System.out.println("Finally block is always executed");
    }
  }
}
```

Output:

```
C:\jdk1.3\bin>javac ExceptionPre.java
C:\jdk1.3\bin>java ExceptionPre

ArithmeticException => / by zero
Finally block is always executed
```

29. Program to illustrate multiple catch blocks

```
import java.io.*;
class ListOfNumbers
{
  public int[] arrayOfNumbers = new int[10];
  public void writeList()
{
    try
{
      arrayOfNumbers[10] = 11;
}
```

```
catch (NumberFormatException e1)
{
     System.out.println("NumberFormatException => " + e1.getMessage());
 }
catch (IndexOutOfBoundsException e2)
{
 System.out.println("IndexOutOfBoundsException => " + e2.getMessage());
 }
 }
}
class MultiCatch
{
  public static void main(String[] args)
 {
    ListOfNumbers list = new ListOfNumbers();
    list.writeList();
  }
}
```

Output:

```
C:\jdk1.3\bin>javac MultiCatch.java
C:\jdk1.3\bin>java MultiCatch
IndexOutOfBoundsException => Index 10 out of bounds for length 10
```

30. Program to illustrate User Defined Exception

```
import java.io.*;
class InvalidAgeException extends Exception
{
      InvalidAgeException(String s)
     {
       super(s);
      }
}
class CustomException
{
     static void validate(int age) throws InvalidAgeException
     {
         if(age<18)
         throw new InvalidAgeException("not valid");
         else
          System.out.println("welcome to vote");
      }
```

```
 public static void main(String args[])
{
try
{
 validate(13);
 }
catch(Exception m){System.out.println("Exception occured: "+m);
}
System.out.println("rest of the code...");
}
}
```

Output:

```
C:\jdk1.3\bin>javac CustomException
C:\jdk1.3\bin>java CustomException

Exception occured: InvalidAgeException: not valid
      rest of the code...
```

5.2 I/O Manipulation Programs

31. Program to read character from keyboard until the user types a "q" −

```
/* This program continues to read and output the same character until we press
'q'*/

import java.io.*;
public class ReadConsole
{
public static void main(String args[]) throws IOException
{
InputStreamReader cin=null;
try
{
cin=new InputStreamReader(System.in);
System.out.println("Enter characters, 'q' to quit.");
char c;
do
{
c =(char)cin.read();
System.out.print(c);
}
while(c !='q');
}
```

```
finally
{
if(cin!=null)
{
cin.close();
}
}
}
}
```

Output:

```
C:\jdk1.3\bin>javac ReadConsole.java
C:\jdk1.3\bin>java ReadConsole
Enter characters, 'q' to quit.
l
l
e
e
q
q
```

32. Program to read bytes from the stream

```
import java.io.*;
public class BISExample
{
public static void main(String[] args) throws Exception
{
byte[] by = new byte[1024];
BufferedInputStream bis = new
        BufferedInputStream(new FileInputStream("BISfile.txt"));
        int i = 0;
        while ((i = bis.read(by)) != -1)
                {
                String st = new String(by, 0, i);
                System.out.print(st);
                }
                bis.close();
        }
}
```

//create a text file “BISfile.txt” and write Hello World, All glitters are not gold.

Output:

```
C:\jdk1.3\bin>javac BISExample.java
C:\jdk1.3\bin>java BISExample
Hello World
All glitters are not gold.
```

33. Program to write text to a file

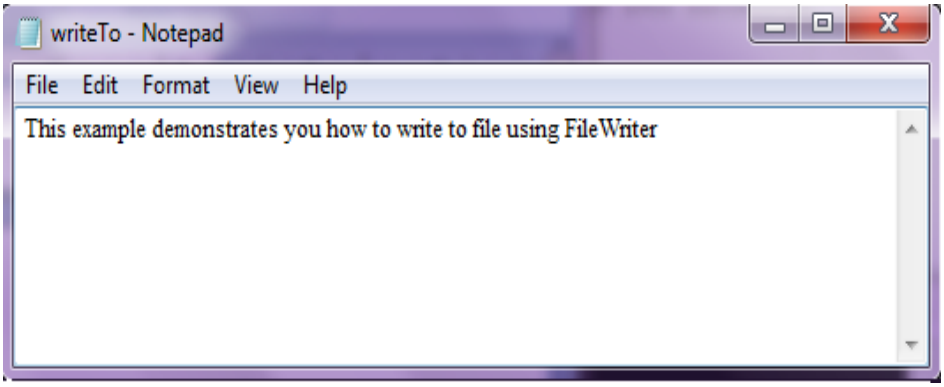
```
import java.io.File;
import java.io.FileWriter;
class WriteToFile
{
public static void main(String args[])
{
String str = "This example demonstrates you how to write to file using
FileWriter";
File file = null;
FileWriter fw = null;
try
{
file = new File("writeTo.txt");
fw = new FileWriter(file);
fw.write(str);
fw.close();
}
catch(Exception e)
{
System.out.println(e);
}
System.out.println("File is created and contents are written into successfully");
}
}
```

Output:

```
C:\jdk1.3\bin>javac WriteToFile.java
C:\jdk1.3\bin>java WriteToFile
```

File is created and contents are written into successfully

When you will execute this example a text file will be created on the specified place as the path given by you with containing the text that you are trying to write in that file by java program like as:

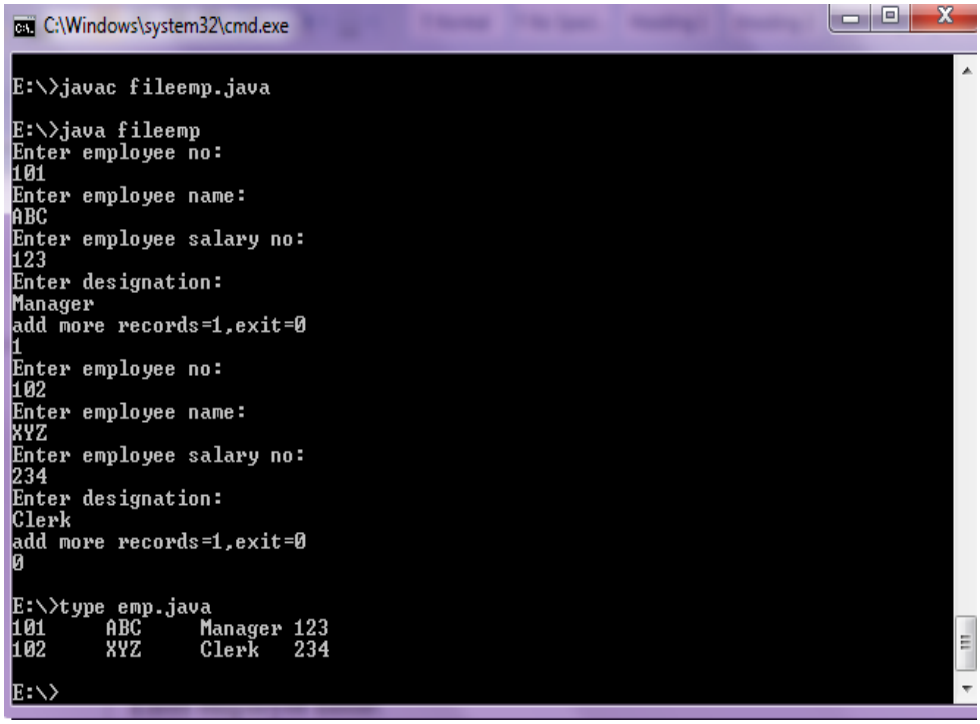


34. Program to demonstrate I/O Operations

```
import java.io.*;
class fileemp
{
public static void main(String args[]) throws IOException
{
int empno,salary;
String name, design;
int more;
FileOutputStreamfos=new FileOutputStream("emp.java");
PrintWriter write=new PrintWriter(fos);
BufferedReader ds=new BufferedReader(new InputStreamReader(System.in));
do
{
System.out.println("Enter employee no: ");
empno=Integer.parseInt(ds.readLine());
System.out.println("Enter employee name: ");
name=(ds.readLine());
System.out.println("Enter employee salary: ");
salary=Integer.parseInt(ds.readLine());
```

```
System.out.println("Enter designation: ");
design=(ds.readLine());
write.println(empno+"\t" +name+ "\t" +design+"\t"+salary);
System.out.println("add more records=1,exit=0");
more=Integer.parseInt(ds.readLine());
}
while(more==1);
write.close();
}
}
```

OUTPUT:



CHAPTER VI

BLOCK V: NETWORK PROGRAMMING

6.1 Applet Programs

35. Simple Applet Program

```
//First.java
import java.applet.Applet;
import java.awt.Graphics;
/*<applet code="First.class" width="200" height="200">
</applet> */
public class First extends Applet
{
public void paint(Graphics g)
{
g.drawString("Welcome",150,150);
}
}
```

Output:

```
C:\jdk1.3\bin>javac First.java
C:\jdk1.3\bin>appletviewer First.java
```

35. Program to draw different shapes using Graphics class

```
import java.io.*;
import java.awt.*;
```

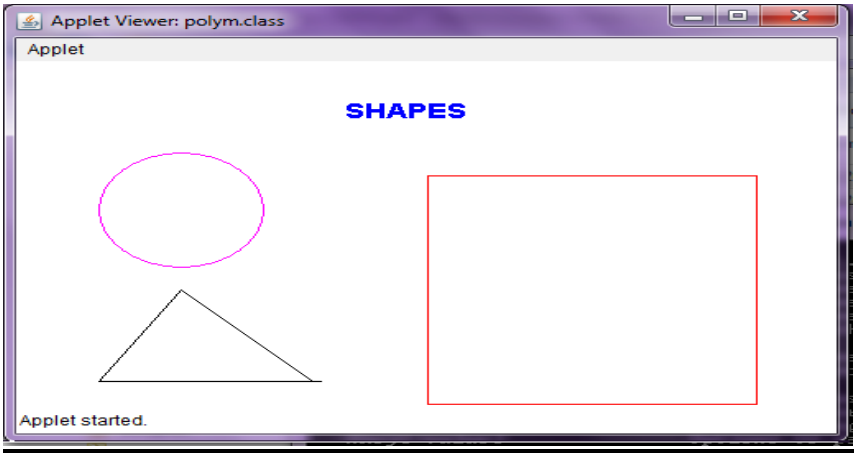
```
import java.applet.*;
/*<applet code="polym.class" width=300 height=80></applet>*/
public class polym extends Applet
{
final Font f=new Font("HELVETICAL" , Font. BOLD,18);
public void paint(Graphics g)
{
g.setColor(Color.BLUE);
g.setFont(f);
draw("SHAPES" ,g);
g.setColor(Color.RED);
draw(250,100,200,200,g);
g.setColor(Color.BLACK);
draw(100,200,g);
g.setColor(Color.MAGENTA);
draw(50,80,100,g);
}
public void draw(String s, Graphics g)
{
g.drawString(s,200,50);
}
public void draw(int x, int y, int m, int n, Graphics g)
{
g.drawRect(x,y,m,n);
}
public void draw(int x1,int y1,Graphics g)
{
g.drawLine(x1,y1,x1-50,y1+80);
g.drawLine(x1-50,y1+80,y1-15,y1+80);
g.drawLine(x1+80,y1+80,x1,y1);
}
public void draw(int x, int y,int r,Graphics g)
{
```

```
g.drawOval(x,y,r,r);
}
}
```

Output:

C:\jdk1.3\bin>javac polym.java

C:\jdk1.3\bin>appletviewer polym.java



36. Program using KeyEvents

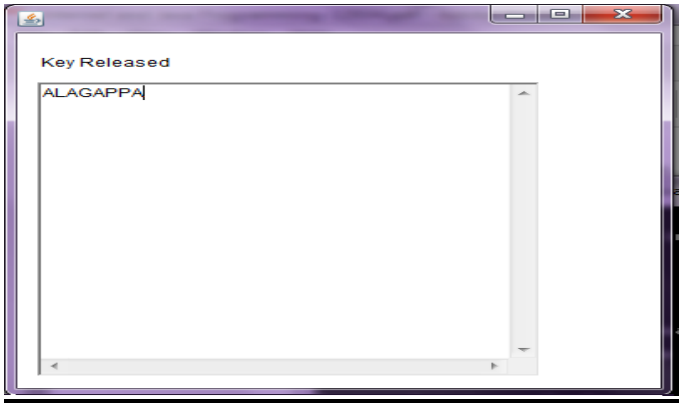
```
import java.awt.*;
import java.awt.event.*;
public class KeyExample extends Frame implements KeyListener
{
   Label l;
   TextArea area;
   KeyExample()
{
      l=new Label();
      l.setBounds(20,50,100,20);
      area=new TextArea();
      area.setBounds(20,80,300, 300);
      area.addKeyListener(this);

      add(l);add(area);
      setSize(400,400);
      setLayout(null);
      setVisible(true);
   }
   public void keyPressed(KeyEvent e)
```

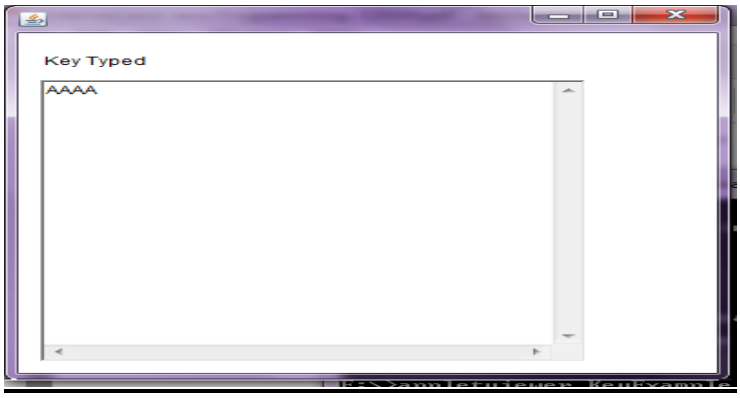
```
 {
      l.setText("Key Pressed");
   }
   public void keyReleased(KeyEvent e)
  {
      l.setText("Key Released");
   }
   public void keyTyped(KeyEvent e)
  {
      l.setText("Key Typed");
   }
     public static void main(String[] args)
  {
      new KeyExample();
   }
}
```

OUTPUT:

C:\jdk1.3\bin>javac KeyExample.java

C:\jdk1.3\bin>java KeyExample





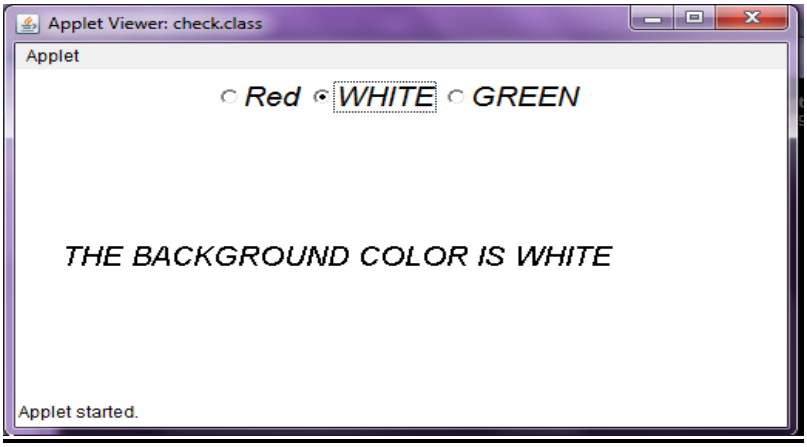
37. Program to demonstrate AWT component - CheckBoxGroup

```
import java.awt.*;
import java.applet.*;
importjava.awt.event.*;
/*<applet code="check.class" width=500 height=250></applet>*/
public class check extends Applet implements ItemListener
{
CheckboxGroup c;
Checkbox b1,b2,b3;
String s;
Font f;
public void init()
{
f=new Font("ARIAL",Font.ITALIC,20);
setFont(f);
s=" ";
c=new CheckboxGroup();
b1=new Checkbox("Red",c,false);
b2=new Checkbox("WHITE",c,false);
b3=new Checkbox("GREEN",c,false);
add(b1);
b1.addItemListener(this);
add(b2);
b2.addItemListener(this);
add(b3);
b3.addItemListener(this);
}
public void start()
{
}
public void stop()
{
```

```
}
public void destroy()
{
}
public void paint(Graphics g)
{
g.drawString(s,30,150);
}
public void itemStateChanged(ItemEvent ie)
{
s=c.getSelectedCheckbox().getLabel();
if(b1.getState())
{
setBackground(Color.RED);
s=("THE BACKGROUND COLOR IS RED");
}
else
if(b2.getState())
{
setBackground(Color.WHITE);
s=("THE BACKGROUND COLOR IS WHITE");
}
else
if(b3.getState())
{
setBackground(Color.GREEN);
s=("THE BACKGROUND COLOR IS GREEN");
}
}
}
```

Output:

```
C:\jdk1.3\bin>javac check.java
C:\jdk1.3\bin>appletviewer check.java
```



6.2 Network Programs

38. Program for Client/Server communication (Socket)

```
//MyClient.java
import java.io.*;
import java.net.*;
public class MyClient {
public static void main(String[] args) {
try{
Socket s=new Socket("localhost",6666);
DataOutputStream dout=new DataOutputStream(s.getOutputStream());
dout.writeUTF("Hello Server");
dout.flush();
dout.close();
s.close();
}catch(Exception e){System.out.println(e);}
}
}

//MyServer.java

import java.io.*;
import java.net.*;
public class MyServer {
public static void main(String[] args){
try{
ServerSocket ss=new ServerSocket(6666);
Socket s=ss.accept();//establishes connection
DataInputStream dis=new DataInputStream(s.getInputStream());
```

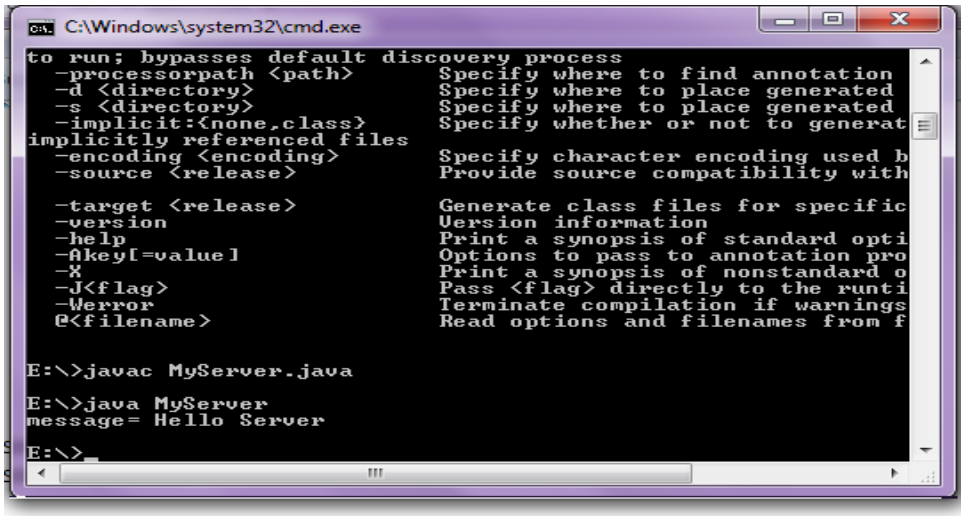
```
String  str=(String)dis.readUTF();
System.out.println("message= "+str);
ss.close();
}catch(Exception e){System.out.println(e);}
}
}
```

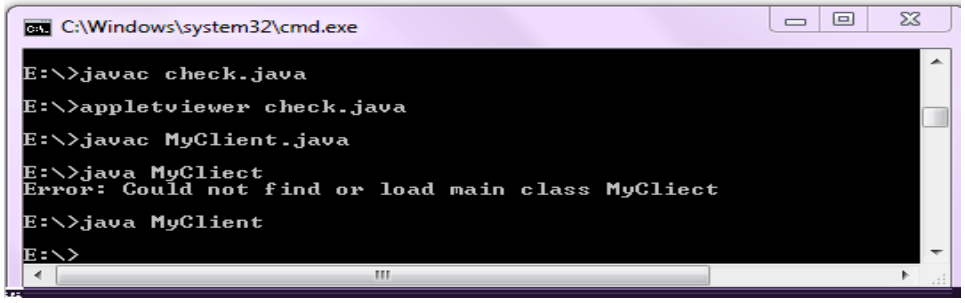
Output:

Open two commad prompt window each for MyClient and MyServer

C:\jdk1.3\bin>javac MyClient.java
C:\jdk1.3\bin>java MyClient

C:\jdk1.3\bin>javac MyServer.java
C:\jdk1.3\bin>java MyServer





37. Program to demonstrate the methods of URL Class

```
//URLDemo.java
import java.net.*;
public class URLDemo{
public static void main(String[] args)
{
try
```

```
{
URL url=new URL("http://www.alagappa.com/java-programs");
System.out.println("Protocol: "+url.getProtocol());
System.out.println("Host Name: "+url.getHost());
System.out.println("Port Number: "+url.getPort());
System.out.println("File Name: "+url.getFile());
}
catch(Exception e){System.out.println(e);
}
}
}
```

OUTPUT:

38. Program for InetAddress class to get IP address of www.javatpoint.com website

```
import java.io.*;
import java.net.*;
public class InetDemo{
public static void main(String[] args){
try{
InetAddress ip=InetAddress.getByName("www.javatpoint.com");
System.out.println("Host Name: "+ip.getHostName());
System.out.println("IP Address: "+ip.getHostAddress());
}catch(Exception e){System.out.println(e);}
}
}
```

Output:

```
C:\jdk1.3\bin>javac InetDemo.java
C:\jdk1.3\bin>java InetDemo

Host Name: www.javatpoint.com
IP Address: 206.51.231.148
```